ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Light: Shadows and Reflections** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** straight lines	Rectilinear propagation.
2	**(d)** a source of light, an opaque object and a screen	All three.
3	**(b)** reflection	Reflection.
4	**(d)** virtual and erect	Cannot be on a screen.
5	**(a)** luminous objects	E.g. the Sun.
6	**(c)** closer to the light source	It blocks more light.
7	**(c)** a mirror	Polished surface.
8	**(a)** same as the object	Equal.
9	**(b)** the Sun	It produces light.
10	**(c)** umbra	Complete shadow.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Luminous objects emit light (1); Non-luminous objects only reflect it (1).	1+1
12	A box with a tiny hole and a screen (1); Forms an inverted image (1).	1+1
13	To see over obstacles (1); Used in submarines (1).	1+1
14	Transparent: light passes fully (1); Translucent partly, opaque not at all (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Look through three aligned cards with holes (1); Light seen when holes are aligned (1); Not when one is shifted (1).	1+1+1
16	Image 2 m behind (1); Total 4 m (1); Equal distances (1).	1+1+1
17	Source is large (1); Complete shadow: umbra (1); Partial shadow: penumbra (1).	1+1+1
	OR: Shadow gets smaller (1); It becomes sharper (1); Source and screen fixed (1).	1+1+1
18	Two mirrors at 45° (1); Light reflected twice (1); Observer sees above obstacle (1).	1+1+1
	OR: It is non-luminous (1); Reflects sunlight (1); Seen by reflection (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Source, object, screen (1); Shadow forms behind (1); Moving the object closer to source enlarges (1); Moving closer to screen shrinks (1); Light travels straight (1).	1+1+1+1+1
	OR: Three mirror strips in a triangle (1); Coloured pieces at one end (1); Multiple reflections (1); Symmetric patterns (1); Rotating changes patterns (1).	1+1+1+1+1
20	Light from top goes down (1); Light from bottom goes up (1); Rays cross at the hole (1); Image inverted (1); Small hole gives sharp image (1).	1+1+1+1+1
	OR: Sun, candle, bulb, stars (1); Moon, book, table (1); We see by reflected light (1); Light enters the eye (1); Need light source (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that the Moon is luminous; it only **reflects** sunlight.
- Writing that the image in a plane mirror is real; it is **virtual** and cannot be caught on a screen.
- Forgetting that the distance between a person and the image is **twice** the distance from the mirror.
- Saying that a transparent object forms a dark shadow; only **opaque** objects form dark shadows.
- Measuring the angles from the mirror surface instead of from the **normal**.
- Writing that a pinhole camera forms an erect image or that a large hole gives a sharper picture.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Light: Shadows and Reflections** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** incident ray	Incident.
2	**(d)** equal to	Equal behind the mirror.
3	**(a)** the Moon	It reflects sunlight.
4	**(a)** penumbra	Partial shadow.
5	**(d)** reflected ray	Reflected.
6	**(b)** interchanged (laterally inverted)	Lateral inversion.
7	**(d)** transparent	E.g. clear glass.
8	**(c)** inverted	Rays cross at the hole.
9	**(b)** normal	Perpendicular.
10	**(b)** it reads correctly in the rear-view mirror	Drivers read it.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Left and right are interchanged (1); In a plane mirror (1).	1+1
12	Equal to incidence (1); 45° (1).	1+1
13	An opaque object blocks light (1); A dark area forms on the screen (1).	1+1
14	Virtual and erect (1); Same size as object (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Box with a small hole on one side (1); Translucent screen on the opposite side (1); Inverted image forms (1).	1+1+1
16	Reversed text appears correct in the mirror (1); Easy to read by drivers (1); Lateral inversion (1).	1+1+1
17	Sun's position changes (1); Long in morning and evening (1); Short at noon (1).	1+1+1
	OR: Inverted (1); Real (1); Size depends on distance (1).	1+1+1
18	Regular: smooth surface, parallel rays (1); Diffused: rough surface, scattered (1); Mirror vs paper (1).	1+1+1
	OR: Smooth water acts like a mirror (1); Images are seen (1); Ripples disturb it (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	i = r (1); Rays and normal in one plane (1); Use a mirror and pins (1); Measure angles (1); Compare (1).	1+1+1+1+1
	OR: Shadow of the Moon on Earth: solar eclipse (1); Shadow of Earth on Moon: lunar eclipse (1); Straight-line travel of light (1); Umbra and penumbra (1); Moon phases due to viewing angle (1).	1+1+1+1+1
20	Virtual and erect (1); Same size (1); 3 m behind (1); Lateral inversion (1); Cannot be on screen (1).	1+1+1+1+1
	OR: Dressing mirror (1); Periscope (1); Kaleidoscope (1); Mirror in vehicles (1); Mirror in dentists (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that the Moon is luminous; it only **reflects** sunlight.
- Writing that the image in a plane mirror is real; it is **virtual** and cannot be caught on a screen.
- Forgetting that the distance between a person and the image is **twice** the distance from the mirror.
- Saying that a transparent object forms a dark shadow; only **opaque** objects form dark shadows.
- Measuring the angles from the mirror surface instead of from the **normal**.
- Writing that a pinhole camera forms an erect image or that a large hole gives a sharper picture.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Light: Shadows and Reflections** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** translucent	E.g. butter paper.
2	**(d)** screen at the back	Translucent screen.
3	**(b)** incidence	Angle i.
4	**(c)** two plane mirrors	At 45 degrees.
5	**(d)** opaque	E.g. wood.
6	**(d)** small	Narrow beam.
7	**(a)** reflection	Angle r.
8	**(b)** multiple reflections	Mirrors inclined.
9	**(c)** an opaque object blocks light	No light reaches the area.
10	**(d)** real	Formed on the screen.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Real: can be taken on a screen (1); Virtual: cannot (1).	1+1
12	Angle of incidence equals angle of reflection (1); Incident ray, normal and reflected ray lie in one plane (1).	1+1
13	Drivers read it correctly in the mirror (1); They can give way (1).	1+1
14	Dressing mirror (1); Periscope or kaleidoscope (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Draw the normal (1); Incident ray at 50° to it (1); Reflected ray at 50° on the other side (1).	1+1+1
16	Wall is rough (1); Diffused reflection (1); Mirror is smooth (1).	1+1+1
17	Submarines (1); Army trenches (1); Crowds (1).	1+1+1
	OR: Virtual and erect (1); Same size (1); Same distance behind (1).	1+1+1
18	Mirrors inclined at an angle (1); Multiple reflections (1); Symmetric patterns (1).	1+1+1
	OR: Angle of incidence = 90 - 30 (1); 60° (1); Angle of reflection 60° (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Tube with two plane mirrors (1); Mirrors parallel at 45° (1); Light reflects twice (1); Eye sees object above (1); Used in submarines (1).	1+1+1+1+1
	OR: Place a pin in front (1); Another pin behind in line with image (1); Parallax method (1); Distances equal (1); Record (1).	1+1+1+1+1
20	Regular reflection: mirror (1); Parallel rays stay parallel (1); Diffused: paper (1); Rays scatter (1); Both obey laws of reflection (1).	1+1+1+1+1
	OR: Glass windows (1); Frosted glass for privacy (1); Wood and metal doors (1); Shadow behaviour (1); Choose by use (1).	1+1+1+1+1

Mistakes to avoid:

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- Forgetting that the distance between a person and the image is **twice** the distance from the mirror.
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